

Microsoft introduces open source static type checker for Python

Microsoft has released Pyright, an open source static type-checking tool for the Python programming language.



It is claimed that Pyright can run five times faster than other existing type checkers written in Python, such as Mypy, Pytype and Pyre.

Microsoft said the tool also runs faster than its own Python Language Server, which offers some overlapping features.

Pyright is written in Typescript and runs on Node.js, which means it does not require a Python environment or imported third party packages to be installed.

It is intended for large Python source bases and can perform fast incremental updates when files are modified.

The tool supports flexible configuration, providing granular control over settings.

Users can set up different execution environments for different subsets of a source base, each of which can use different module search paths, Python language versions and target platforms.

Pyright supports features such as type hints, variable notation and structural subtyping.

It can be used as a Language Server Protocol plugin for Microsoft's Visual Studio Code source-code editor or can be run as a standalone command-line tool, Microsoft sources said.

Pyright currently provides support for Python 3.0 and its newer versions only. Microsoft said it has no plans to support older versions as of now.

Sources at Microsoft added, "Pyright is a work in progress. Type-checking capabilities are not fully implemented."

A non-profit industry organisation composed of the industry's leading broadband operators, vendors and thought leaders, Broadband Forum is focused on accelerating broadband innovation, standards and ecosystem development.

Created as part of Broadband Forum's Open Broadband initiative, the Open Broadband – USP Agent (OB-USP-Agent) project will provide vendors with a code base that they can either integrate into their devices or use as a reference implementation as they use USP. A necessary catalyst to adoption in today's

software development world, OB-USP-Agent will facilitate USP deployment and result in faster time-to-market for USP based solutions and innovation.

Thanks to this standards based solution, operators now no longer have to depend on costlier and less flexible proprietary

management solutions. Published in April 2018, USP is an evolution of the TR-069 standard, currently installed in more than one billion households globally.

Using the same data models as TR-069, USP builds a network of controllers and agents to allow applications to manipulate service elements. This allows service providers, consumer electronics manufacturers and end users to securely manage connected devices, and carry out upgrades, for example, for critical security updates, as well as onboard new devices. Customer support is also improved by remote monitoring and troubleshooting of connected devices, services and home network links.

"USP is designed to be flexible, scalable and secure, and the new open source agent implementation builds on this by allowing vendors to integrate it into their devices or use it as a reference implementation," said Barbara Stark, of AT&T, USP Project Lead at Broadband Forum.

"This is a natural evolution for USP as the industry moves more and more towards software development. We are confident the new agent will not only help to increase interoperability but also support operators looking to launch new connected home services as additional revenue streams," Stark added.

The open source USP Agent has been provided by Broadband Forum member ARRIS International plc, which has already used the agent in previous USP Plugfest events and tested it against the existing USP Compliance Test Plan.

"OB-USP-Agent marks an important step in the evolution of USP, aiding adoption rates on devices," said John Blackford, product management director at ARRIS and co-director of the Broadband User Services (BUS) Work Area.

He added, "Through the vast number of TR-069 deployments, USP is the result of the evolution of a platform that already has a proven track record and we are confident it will support service providers looking to capitalise on the IoT market by enabling an open market without vendor lock-in."

Project participants can leverage and extend this code as innovation around OB-USP-Agent grows. The project has been launched and the code is available for Broadband Forum members. A public release will be made available later this year.



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